DSC 478 - Programming Machine Learning Applications

Course Syllabus

INSTRUCTOR

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Office Hours: Thu 4:00-5:30 PM (**appointments required via OneDePaul**)

COURSE DESCRIPTION

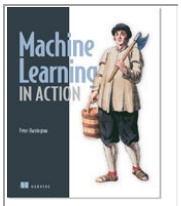
The course will focus on the implementations of various data mining and machine learning techniques and their applications in various domains. The primary tools used in the class are the Python programming language and several associated libraries. Additional open-source machine learning and data mining tools may also be used as part of the class material and assignments. Students will develop hands on experience developing supervised and unsupervised machine learning algorithms and will learn how to employ these techniques in the context of popular applications such as automatic classification, recommender systems, searching and ranking, text mining, group and community discovery, and social media analytics.

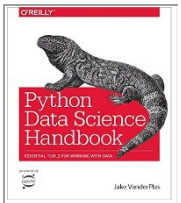
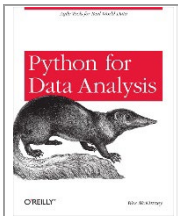
PREREQUISITES

(DSC 441 and DSC 430) or CSC 480

TEXTBOOKS & COURSE MATERIAL

We will use numerous online resources and documents throughout the course. The required and recommended textbooks are listed below. Other resources and course materials directly relevant to topics covered in the course will be available in the Content section of the D2L site for the course.

Required Text	
	**Machine Learning in Action**, by Peter Harrington, Manning Publications.

Recommended Texts	
	**Python Data Science Handbook: Essential Tools for Working with Data**, by Jake VanderPlas, Second Edition, 2023.
	**Python for Data Analysis**, by Wes McKinney, 3nd Edition, O'Reilly.

GRADING & COURSE REQUIREMENTS

The structure and grading in the class will be centered around 4 assignments and a final project. The assignments will involve Python implementations of selected machine learning techniques and their applications in various domains. They will also involve the use of various Python libraries to perform preprocessing, data exploration/visualization, and data analysis using different data sets. These assignments must be done individually, unless otherwise specified. You may discuss the assignments with others in the class, but you must develop your own solution to the problems in the assignment. Late assignments will be penalized 10% per day (with weekends counting as one day).

The final project will involve either developing and evaluating an application that uses one or machine learning algorithms to perform a specific task; or they may involve performing a complete analysis of a complex data set using Python tools. The goal of the project is to integrate several concepts covered during the quarter to achieve a more complex task than those explored in the assignments. Students can propose their own project idea (to be approved) and may complete the project either individually or in groups of up to three members. More details on the final project will be available in the Content section of the D2L site for the course.

The final grade will be determined (tentatively) based on the following components:

Assignments = 65%
Final Project = 35%

The general grading scheme will be based on a curve. At the end of the quarter, some adjustments may be made based on overall class performance as well as signs of individual effort. Pluses and minuses will be given at the high/low ends of each grade range.

TENTATIVE LIST OF TOPICS

The following issues and topics will be covered throughout the course. Many of these topics will be revisited several times during the course in a variety of contexts.

- **The Knowledge Discovery Process and Machine Learning Overview**
  - The KDD process and methodology
  - Data preparation for knowledge discovery and machine learning
  - Exploratory data analysis
  - Overview of machine learning tasks
  - Review of Python and overview of Python tools for data analysis and visualization
- **Supervised Learning**
  - Classification and prediction using K-Nearest-Neighbor
  - Probabilistic models: Naïve Bayes
  - Building decision trees
  - Linear regression models and regularization
  - Gradient descent optimization
  - Support vector machines
  - Evaluating predictive models
  - Hyper-parameter tuning
  - Ensemble models: bagging and boosting, including Random Forest, AdaBoost, etc.
  - Artificial Neural Networks
- **Unsupervised Learning**
  - Clustering: K-Means, HSCAN, etc.
  - Hierarchical clustering algorithms
  - Principal Component Analysis and dimensionality reduction
  - Singular Value Decomposition
  - Matrix factorization
- **Possible Applications (covered throughout the course)**
  - Recommender systems and personalization
  - Document categorization
  - Concept discovery from text
  - Building predictive models for target marketing
  - Image segmentation

COURSE AND UNIVERSITY POLICIES

Attendance/Participation: It is expected that you will review all the weekly material for the classes in a timely manner, including lecture videos, class examples, and assigned readings. You should also regularly participate in the discussion forums and post questions and/or suggestions to help other members of the class. During the first week, you are required to post an introductory message about yourself on the discussion forum.

Academic Integrity Policy: This course will be subject to the academic integrity policy of the university. More information can be found at **http://academicintegrity.depaul.edu/**. The university and school policy on plagiarism can be summarized as follows: Students in this course should be aware of the strong sanctions that can be imposed against someone guilty of plagiarism. If proven, a charge of plagiarism could result in an automatic F in the course and possible expulsion. The strongest sanctions will be imposed on anyone who submits as their own work any assignment or part thereof which has been developed or prepared by someone else. Any use of AI tools for preparing any part of the submitted assignments must be acknowledged in detail. If you have any questions or doubt about what plagiarism entails or how to properly acknowledge source materials, be sure to review the policy and consult the instructor.

Incompletes: An incomplete grade is given only for an exceptional reason such as a death in the family, a serious illness, etc. Any such reason must be documented. Any incomplete request must be made at least two weeks before the finals week and approved by the Dean of the College of Computing and Digital Media. Poor performance in the course or missing assignments during the quarter due to negligence will not be considered as valid reasons for such a request.

Resources for Students with Disabilities: Students who feel they may need accommodation based on the impact of a disability should register with DePaul's Center for Students with Disabilities (CSD) enabling them to access accommodation and support services to assist with their success. There are two office locations: Loop Campus – Lewis Center #1420 – (312) 362-8002, and Lincoln Park Campus – Student Center #370 – (773) 325-1677. This is best done early in the term. The conversation will remain confidential to the extent possible. Please see:

https://offices.depaul.edu/student-affairs/about/departments/Pages/csd.aspx

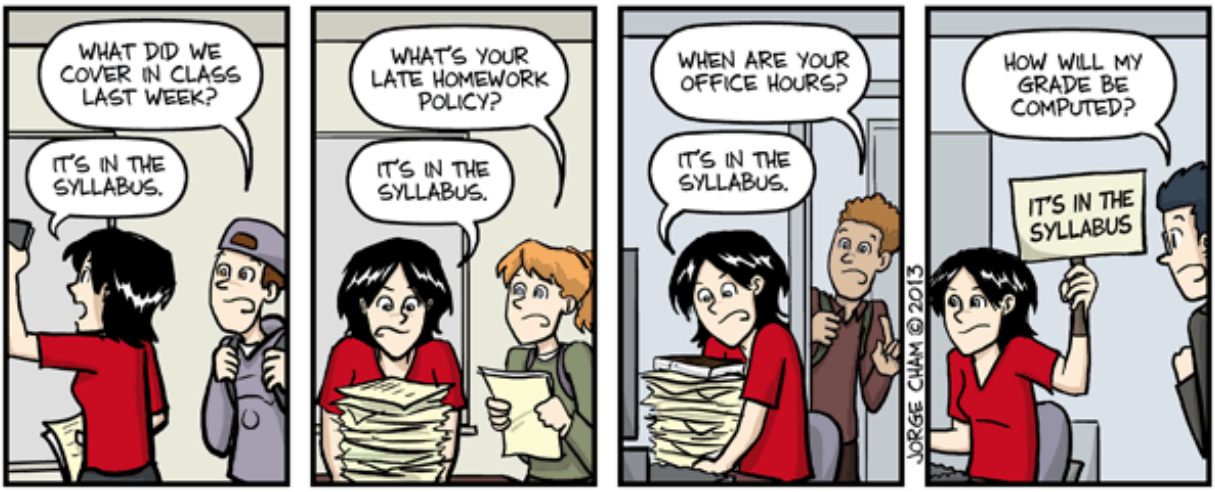
for Services and Contact Information.

Online Teaching Evaluation: Evaluations are a way for students to provide valuable feedback regarding their instructor and the course. Detailed feedback will enable the instructor to continuously tailor teaching methods and course content to meet the learning goals of the course and the academic needs of the students. The evaluations are anonymous; the instructor and administration do not track who entered what responses. A program is used to check if the student completed the evaluations, but the evaluation is completely separate from the student's identity. Periodic reminders are sent to students over the last three weeks of the quarter to complete the online evaluation. Students do not receive reminders once they complete the evaluation. Students complete the evaluation online in Campus Connect.

Tentative Weekly Schedule

	Topic	Assignments & Readings
Week 1	• Introduction to the Course • Overview of Machine Learning and Knowledge Discovery Process • Review of Numpy, Pandas, and Jupyter Notebooks.	• Install and test Python distribution from **Anaconda** which automatically installs all of the libraries used in this class). • Familiarize yourself with IPython, and particularly, **Jupyter Notebook**. See also this **Jupyter Notebook Tutorial Video** by Corey Schafer. • Go through the "Quick Tutorial" on **Numpy User Guide** and try to follow the examples on your own (using Jupyter Notebook as the shell). • Review Section 1 of **Python Scientific Lecture Notes**.
Week 2	• Understanding the Characteristics of Data • Data Preparation and Preprocessing • Using Pandas for Data Preparation and Preprocessing	• Familiarize yourself with Pandas basics. A good place to start is the **Pandas Tutorials** page in Pandas Documentation. You might also review **Python Pandas Tutorial: A Complete Introduction for Beginners.** • Read Section 1.4 of the **Python Scientific Lecture Notes on Matplotlib**
Week 3	**Supervised Learning** • Distances, Similarities, and K-Nearest-Neighbor Search • Classification & Prediction - Review of Basic Concepts	• Read **Chapter 2** of **Machine Learning in Action (MLA)**
Week 4	**Supervised Learning** • Review of Decision Tree Classification • Bayesian Classification • Application Focus: Text Categorization	• Read **Chapters 3** and **4** of **MLA** • Read **scikit-learn user guide**: Sections: **1.2, 1.6, 1.9, 1.10**
Week 5	**Supervised Learning** • Classification (Continued) • Review of Linear Models for Prediction • Application Focus: Recommender Systems	• Read **Chapter 8** of **MLA** • Review **scikit-learn user guide**: Sections: **1.1, 3.1**. • Read **Recommender Systems** Article in the Encyclopedia of Machine Learning • Read Wikipedia article on **Collaborative Filtering**

Week 6	**Supervised Learning** • Review of Multiple Linear Regression and Regularization • Gradient Descent Optimization • Feature Selection, Model Selection, and Hyper-Parameter Optimization	• Read **Chapter 10** of **MLA** • Review **scikit-learn user guide**: Sections: **1.1, 1.5, 3.1, 3.4.**
Week 7	**Unsupervised Learning** • Clustering • Application Focus: Document Clustering	• Read **Chapters 13 of MLA** • Review **scikit-learn user guide**: Section: **2.3** • Review **Wikipedia** pages on **Cluster Analysis**, including the article on **Kmeans Clustering** and **Hierarchical Clustering**
Week 8	**Unsupervised Learning** • Principal Component Analysis (PCA) • Singular Value Decomposition (SVD) and Matrix Factorization	• Read **Chapters 11 and 14** of **MLA** • Read: **Matrix Factorization: A Simple Tutorial and Implementation in Python**, by Albert Au Yeung • Review **scikit-learn user guide**: Section: **2.5.**
Week 9	• Support Vector Machines • More on Hyper-Parameter Tuning • Using Machine Learning Pipelines in Scikit-learn	• Review **scikit-learn user guide**: Sections: **1.4, 3.2, 7.1.** • **Support Vector Machines:** YouTube Video by StatQuest.
Week 10	• Ensemble Classification Methods • Overview of Neural Networks for Supervised Learning	• Review **scikit-learn user guide**: Sections: **1.11, 1.17.** • Wikipedia page on **Artificial Neural Networks**



IT'S IN THE SYLLABUS

This message brought to you by every instructor that ever lived.

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